LANDGUARD AI

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LANDGUARD AI

(PROJECT OVERVIEW)

LandGuard AI is a proposed geospatial early-warning and landslide risk monitoring system tailored specifically for the vulnerable terrains of the North Eastern Region of India. This project aims to integrate environmental, satellite, and rainfall data with interpretable machine learning techniques to estimate both permanent slope susceptibility and immediate rainfall-triggered risks. The primary objective of this system is to transition from reactive disaster response to a proactive, data-driven early warning mechanism that can assist local authorities in identifying highly unstable slopes before catastrophic failures occur.

PROBLEM IDENTIFICATION

(THE VULNERABLITY OF NORTH EASTERN ZONE)

The North Eastern Region of India features highly fragile geological terrain, making it exceptionally vulnerable to frequent and severe landslides. These catastrophic events are primarily triggered by intense seasonal rainfall, natural slope instability, and human-induced ground disturbance. The resulting slope failures consistently lead to blocked transportation networks, destroyed infrastructure, and completely isolated local communities. Currently, monitoring these vast and inaccessible hilly areas relies heavily on manual observation or localized physical sensors, which are difficult to scale geographically and often fail to provide sufficient early warning before a slope completely collapses.

The increasing frequency and impact of landslides demand timely and proactive risk detection. A scalable early-warning system is urgently needed to identify vulnerable slopes before failure, enabling authorities to take preventive action, minimize infrastructure damage, and protect isolated communities.

Literature Review

Current Approaches in Landslide Prediction

Remote Sensing & Explainable Susceptibility

Youssef et al. [1] (2023) developed an interpretable neural-network framework for landslide susceptibility in the eastern Himalaya. The model achieved performance comparable to deep neural networks while identifying slope–precipitation interaction and aspect as major susceptibility factors, demonstrating the value of interpretable ML for landslide assessment.

Multi-Sensor Satellite Analysis

Fu et al. [2] (2024) proposed a landslide-dating framework combining Sentinel-1 SAR and Sentinel-2 optical imagery. The combination of surface-variation and vegetation-change information improved landslide-date detection compared with either sensor alone, demonstrating the benefit of multi-source satellite observations.

Dynamic Rainfall-Based Prediction

Kang et al. [3] (2024) developed a machine-learning forecasting and early-warning model using terrain factors, historical rainfall, and current rainfall. The study demonstrates that combining persistent environmental conditions with rainfall information can improve rainfall-induced landslide forecasting.

Rainfall Threshold Modeling

Menon and Kolathayar [4] (2025) investigated rainfall thresholds using historical landslide events and multiple ML classifiers, including KNN, Random Forest, GBM, SVM, and Logistic Regression. The study demonstrates the importance of antecedent rainfall and rainfall-duration relationships for early warning.

Key Finding

Current research successfully addresses individual components such as susceptibility mapping, satellite-based detection, rainfall thresholds, and explainable ML, but these capabilities remain largely studied as separate modeling tasks.

Research Gap

The need for Dynamic and Explainable Risk Integration

1. Static vs. Dynamic Risk

Many susceptibility models primarily estimate where landslides are likely to occur, but a susceptibility map does not necessarily indicate when a vulnerable slope is becoming dangerous.

2. Fragmented Environmental Information

Satellite observations, terrain characteristics, historical landslide inventories, and rainfall conditions are often modeled separately or within individual prediction frameworks. This limits the ability to continuously combine persistent susceptibility with short-term triggering conditions.

3. Limited Operational Context

A risk probability alone does not provide sufficient information for disaster-management decisions. Authorities need to know which factors are driving the current risk, such as accumulated rainfall, slope characteristics, vegetation degradation, or ground deformation.

4. Explainability–Prediction Gap

Recent work has demonstrated that interpretable ML can provide meaningful

susceptibility explanations [1], while rainfall-based studies demonstrate the importance of dynamic triggering conditions [3], [4]. However, a unified framework that combines both functions for location-specific operational warning remains an important opportunity.

Research Gap

The key gap is the lack of a unified, dynamic, and explainable framework that separates inherent terrain susceptibility from immediate rainfall-triggered risk and converts both into actionable warnings.

Novelty And Original Research Contribution

Our primary research contribution is the development of a Two-Stage Hybrid Risk Architecture coupled with an Explainable Artificial Intelligence (XAI) layer. Instead of processing all environmental variables simultaneously in a single opaque algorithm, our methodology structurally separates inherent landscape vulnerability from immediate meteorological triggers. The first stage establishes a foundational terrain susceptibility score using relatively stable topographical and satellite features, while the second stage dynamically modulates this score using antecedent and current rainfall data. Furthermore, the integration of an explainability framework ensures that every risk prediction is accompanied by a transparent, factor-wise breakdown, transforming raw probabilistic outputs into interpretable, actionable advisories.

LandGuard AI does not treat landslide risk as a single static prediction problem. It structurally separates “where a slope is inherently vulnerable” from “what is currently triggering failure,” then combines both through an explainable risk layer.

Methodology & System Design

Data Acquisition & Preprocessing:

LandGuard AI integrates SRTM terrain data, Sentinel-1 SAR, Sentinel-2 imagery, rainfall time series, antecedent rainfall metrics, and historical landslide inventories. The data are spatially aligned, cleaned, normalized, and transformed into a unified feature dataset.

Two-Stage Risk Modeling:

In Stage 1, Random Forest or Gradient Boosting models analyze relatively stable terrain and satellite features to generate a baseline terrain susceptibility score. In Stage 2, this score is dynamically combined with current and antecedent rainfall conditions to estimate the immediate probability of landslide occurrence.

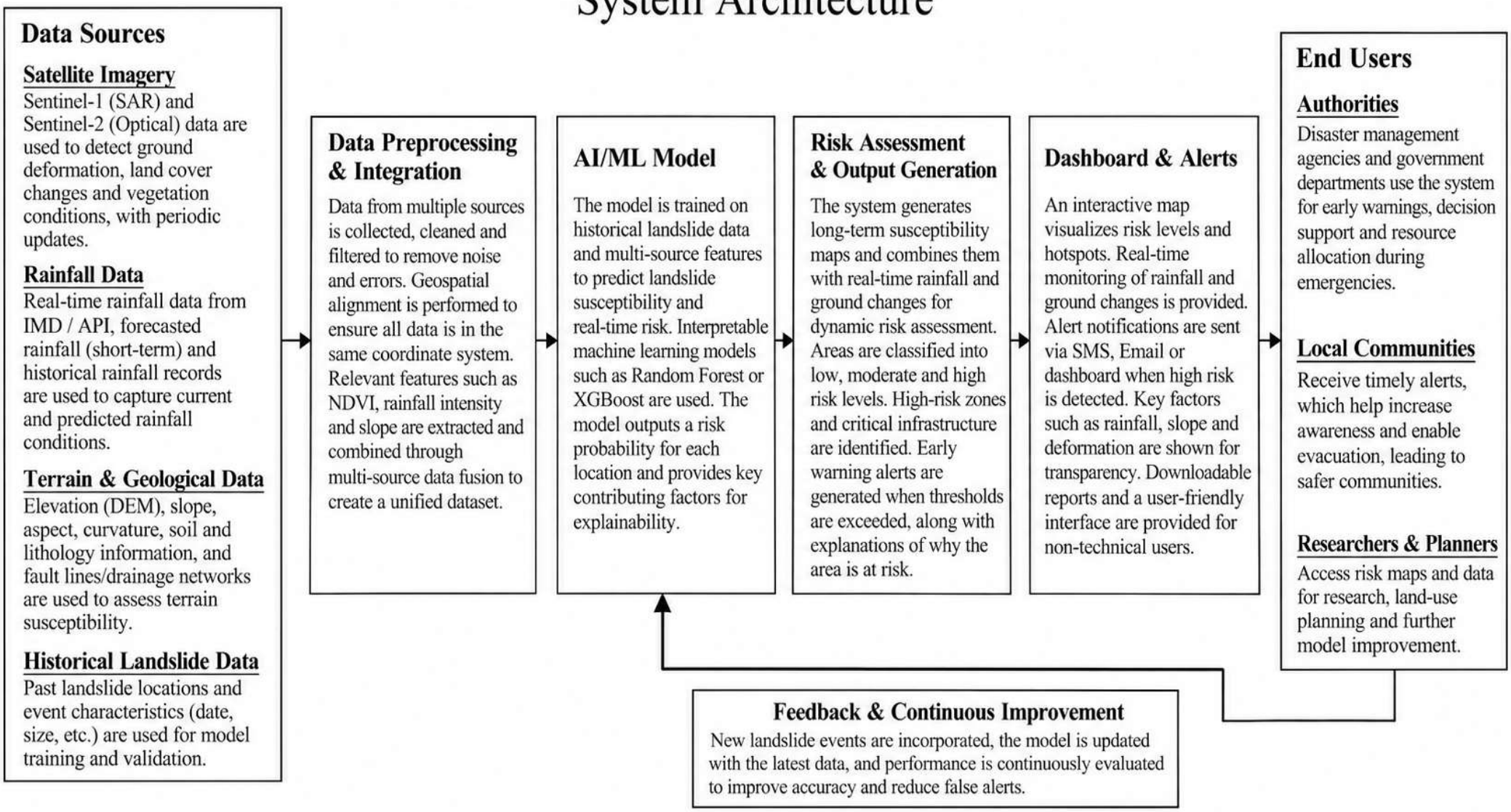
Explainable Risk Assessment:

The final prediction is processed through an XAI layer using SHAP or feature-importance analysis, identifying the contribution of factors such as slope, rainfall, vegetation change, and ground disturbance. This converts model predictions into interpretable, factor-wise risk advisories.

System Design & Deployment:
A Node.js backend manages data pipelines, model execution, and APIs, while a React-based geospatial dashboard provides interactive risk maps, rainfall information, alerts, and XAI explanations. Field reports can be submitted through the platform to support localized validation and continuous model improvement.

Validation:
The system is evaluated against a single-stage ML baseline using historical landslide events, with Precision, Recall, F1-score, and spatial cross-validation used to assess predictive performance and generalization.

Architecture Diagram/ Block Diagram



Expected Impact and Future Scope

The successful implementation of LandGuard AI is expected to deliver a scalable, transparent early-warning prototype that significantly enhances the proactive capabilities of disaster management authorities in the North Eastern Region. By providing explainable, location-specific risk assessments, the system bridges the critical gap between raw environmental data and actionable on-ground response. Looking beyond the immediate scope of rainfall-induced landslides, the system's decoupled, modular architecture is explicitly designed for future extensibility. As part of long-term future work, the core predictive engine and geospatial dashboard can be systematically adapted to incorporate specialized data layers and modeling techniques for other localized environmental hazards, such as flash floods or forest fires, eventually evolving into a comprehensive multi-disaster intelligence platform.

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